

$$\begin{array}{lll}
 15 = 151's, 30's & 16 = 161's, 20's & 17 = 171's, 10's & 18 = 181's \\
 19 = 116's, 31's, 140's & 20 = 116's, 41's, 130's & 21 = 116's, 51's, 120's \\
 22 = 116's, 61's, 110's & 23 = 116's, 71's, 100's & 24 = 116's, 81's, 90's \\
 25 = 116's, 91's, 80's & 26 = 116's, 101's, 70's & 27 = 116's, 111's, 60's \\
 28 = 116's, 121's, 50's & 29 = 116's, 131's, 40's & 30 = 116's, 141's, 30's \\
 31 = 116's, 151's, 20's & 32 = 116's, 161's, 10's & 33 = 116's, 171's \\
 34 = 216's, 21's, 140's & 35 = 216's, 31's, 130's & 36 = 216's, 41's, 120's \\
 37 = 216's, 51's, 110's & 38 = 216's, 61's, 100's & 39 = 216's, 71's, 90's \\
 40 = 216's, 81's, 80's & 41 = 216's, 91's, 70's & 42 = 216's, 101's, 60's \\
 43 = 216's, 111's, 50's & 44 = 216's, 121's, 40's & 45 = 216's, 131's, 30's \\
 46 = 216's, 141's, 20's & 47 = 216's, 151's, 10's & 48 = 216's, 161's \\
 49 = 316's, 11's, 140's & 50 = 316's, 21's, 130's & 51 = 316's, 31's, 120's \\
 52 = 316's, 41's, 110's & 53 = 316's, 51's, 100's & 54 = 316's, 61's, 90's \\
 55 = 316's, 71's, 80's & 56 = 316's, 81's, 70's & 57 = 316's, 91's, 60's \\
 58 = 316's, 101's, 50's & 59 = 316's, 111's, 40's & 60 = 316's, 121's, 30's \\
 61 = 316's, 131's, 20's & 62 = 316's, 141's, 10's & 63 = 316's, 151's \\
 64 = 416's, 140's & 65 = 416's, 11's, 130's & 66 = 416's, 21's, 120's \\
 67 = 416's, 31's, 110's & 68 = 416's, 41's, 100's & 69 = 416's, 51's, 90's \\
 70 = 416's, 61's, 80's & 71 = 416's, 71's, 70's & 72 = 416's, 81's, 60's \\
 73 = 416's, 91's, 50's & 74 = 416's, 101's, 40's & 75 = 416's, 111's, 30's \\
 76 = 416's, 121's, 20's & 77 = 416's, 131's, 10's & 78 = 416's, 141's. \square
 \end{array}$$

However, if we decided this was enough checking, we would come to the wrong conclusion. It is not true that every positive integer is the sum of 18 fourth powers because 79 is not the sum of 18 fourth powers.

$$\begin{array}{l}
 3^4 = 81 > 79. \text{ So, we will have to restrict ourselves with } 2^4 = 16, 1^4 = 1 \\
 \text{and } 0^4 = 0. \quad 16 \times 5 = 80 > 79 \quad 79 = 791's \rightarrow 79 \\
 \quad \quad \quad \quad \quad \quad \quad \quad 79 = 416's, 151's \rightarrow 19.
 \end{array}$$